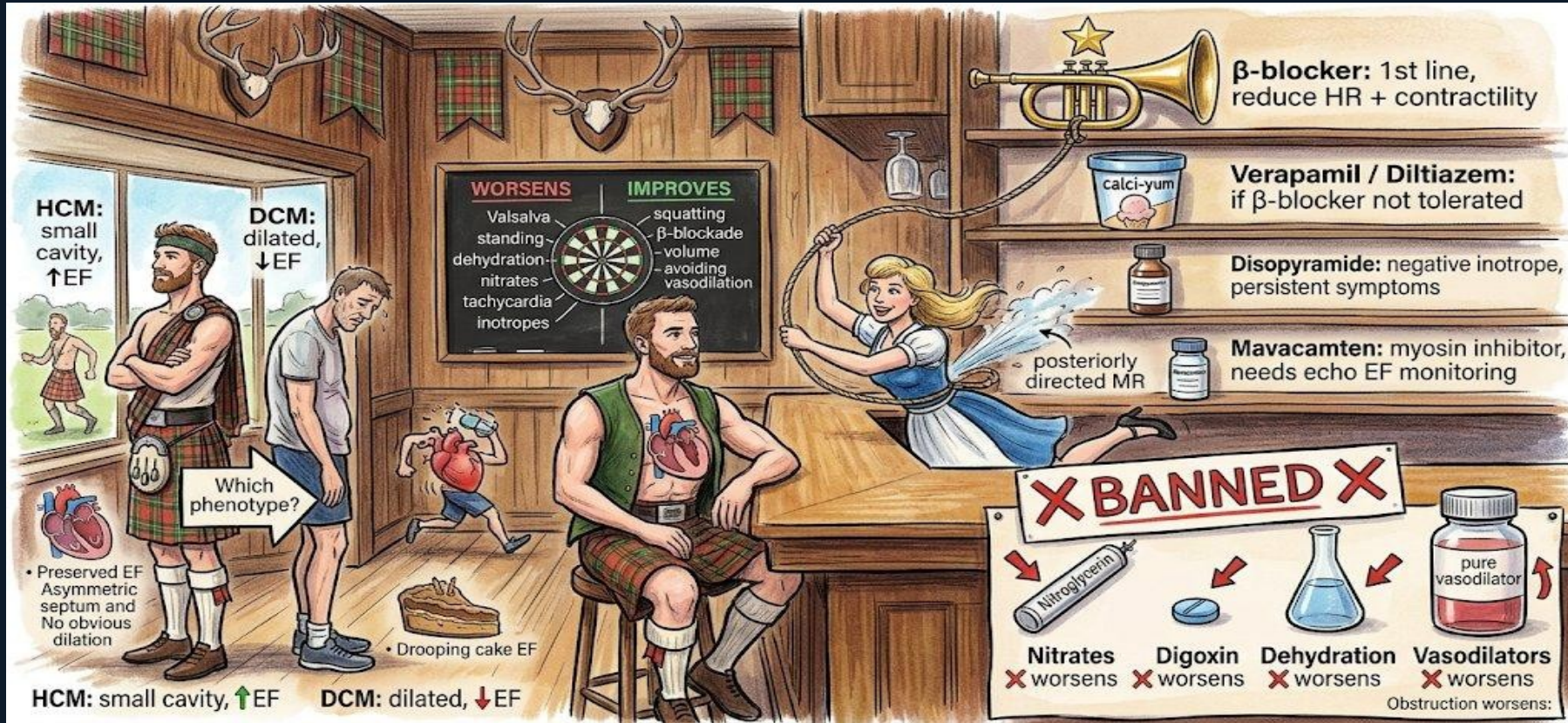


Scene B — HCM: Obstruction & Treatment

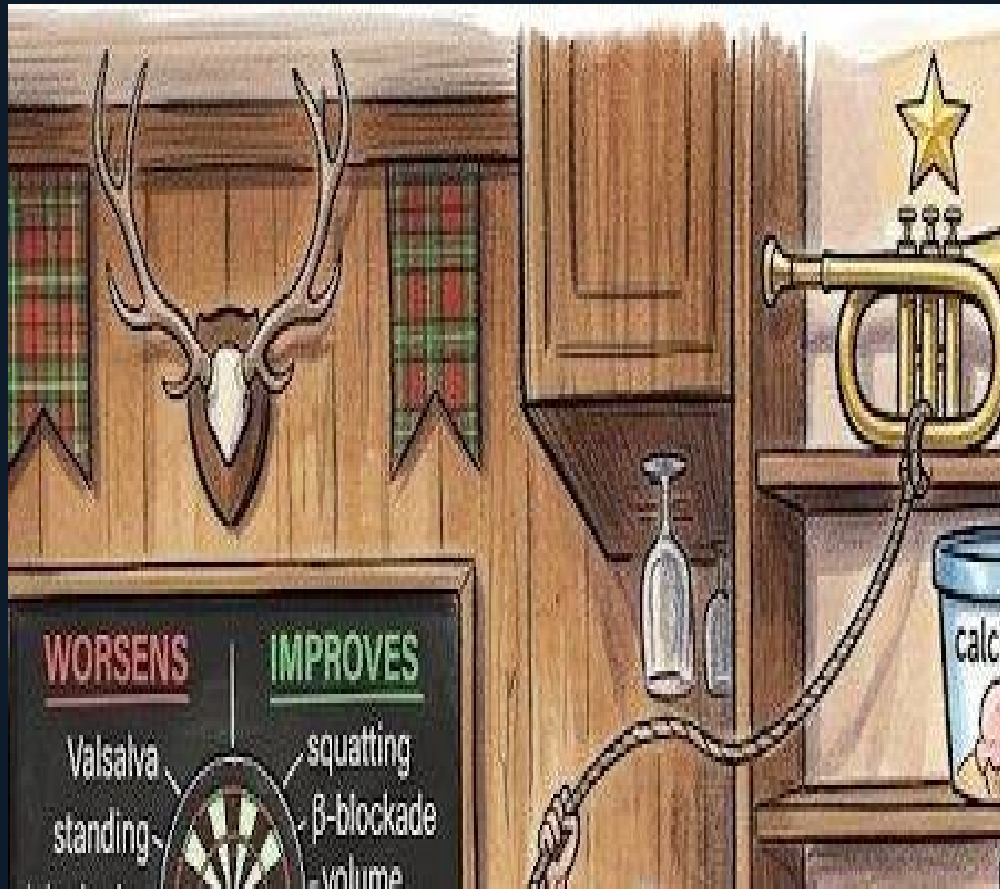


Scottish Pub · 5 board questions



SAM: The Barmaid Swings

The barmaid on a rope swinging toward the thick septum wall is systolic anterior motion (SAM) of the anterior mitral leaflet. During systole, the enlarged anterior leaflet is pulled toward the hypertrophied septum by a Venturi effect from rapid early ejection. When it contacts the septum, it obstructs the LVOT. The posteriorly directed MR spray is the simultaneous secondary consequence — the valve closes incompletely, creating a posteriorly directed regurgitant jet. SAM → LVOT obstruction AND posteriorly directed MR occur together.

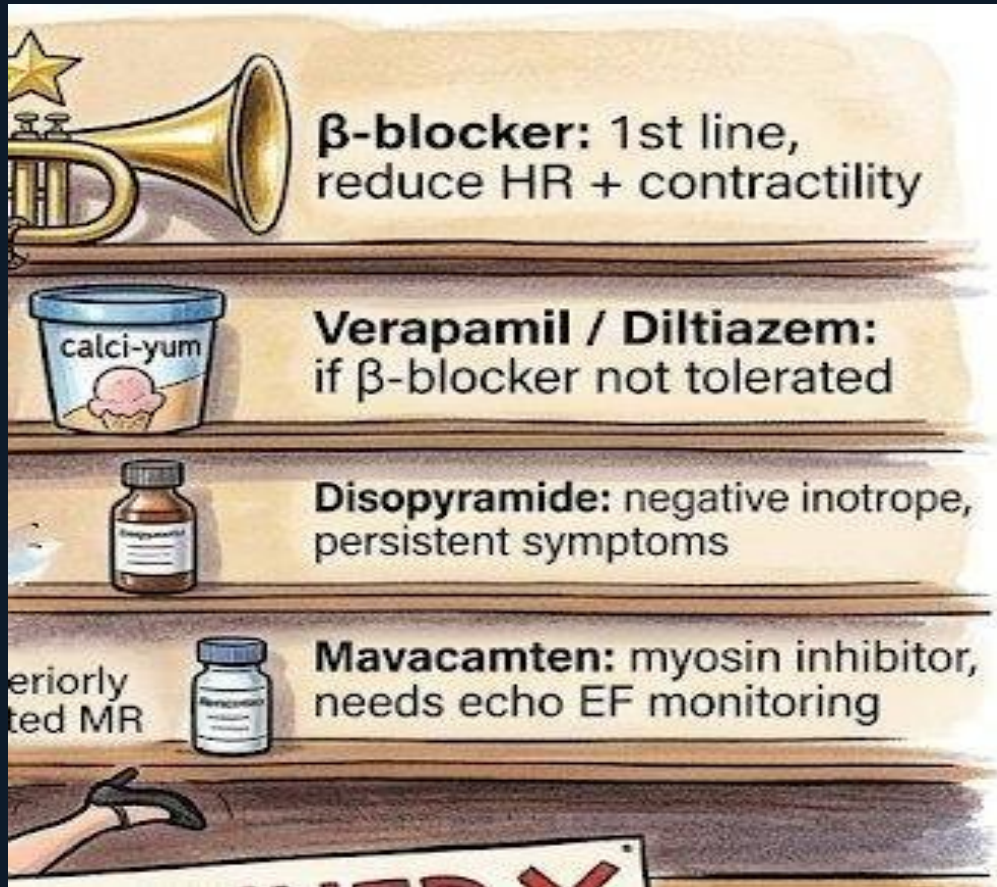


Scene B — HCM: Obstruction & Treatment | Scottish Pub

Worsens vs Improves Dartboard

WORSENS obstruction (red half): Valsalva, standing, dehydration, nitrates, tachycardia, positive inotropes.

IMPROVES obstruction (green half): squatting, beta-blockade, volume loading, avoiding vasodilation. The mechanism is always the same: HCM obstruction worsens when the LV cavity shrinks. Anything reducing preload or increasing contractility shrinks the cavity. Anything increasing preload or reducing contractility expands it. This single rule generates all 10 items on the dartboard.



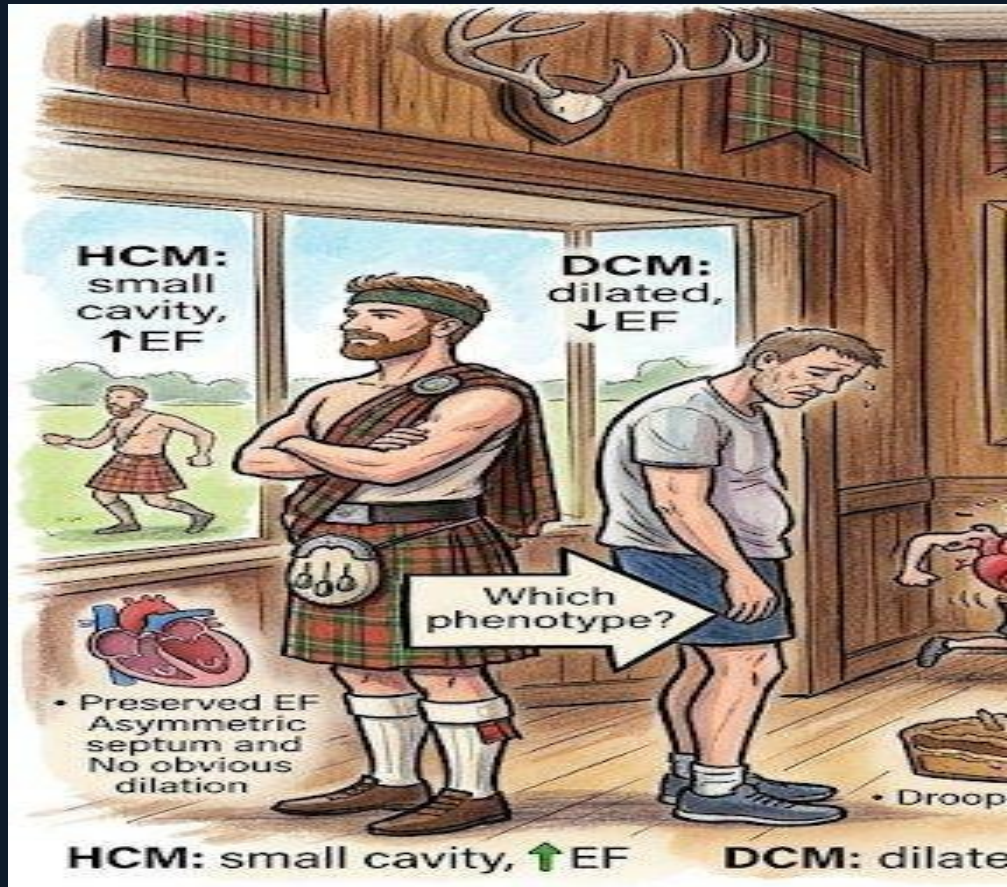
Treatment Shelf: 4-Drug Hierarchy

Beta-blocker bugle with gold star = first-line (reduces HR and contractility, preserves preload). Calci-yum ice cream (verapamil/diltiazem) = second if beta-blocker not tolerated — same dual mechanism. Disopyramide = third: potent negative inotrope for persistent obstruction. Mavacamten = fourth: cardiac myosin inhibitor, needs echo EF monitoring. Escalate in this exact order before invasive septal reduction. The board tests which drug comes first and what is used when beta-blockers fail.



The BANNED Counter

Nitrates: reduce preload $\rightarrow$ smaller LV $\rightarrow$ more obstruction. Digoxin: positive inotrope $\rightarrow$ more powerful systolic ejection $\rightarrow$ more SAM. Dehydration: reduces preload $\rightarrow$ same as nitrates. Pure vasodilators: drop afterload $\rightarrow$ LV empties more aggressively $\rightarrow$ worse gradient. Board trap: HCM patient with angina — 'give nitroglycerin.' Wrong. Use beta-blockers for angina in HCM. Nitrates are contraindicated because they worsen the very obstruction causing the angina (through demand, not supply).



HCM vs DCM: Phenotype First

The window shows the mandatory phenotype classification. HCM: small or normal LV cavity, preserved or hyperdynamic EF, asymmetric septal hypertrophy — the problem is too much wall, not a weak pump. DCM: dilated LV, reduced EF — a weak pump. These two entities have completely opposite treatment philosophies: HCM treatment reduces contractility and preserves preload; DCM treatment increases contractility and afterload reduction. Classifying the phenotype correctly is the first clinical step.

Scene B — Quick Reference

SAM = LVOT obstruction + posterior MR

Anterior leaflet pulls to septum → dual consequence

Improves: squatting, β -blockade, volume

↑ preload or ↓ contractility → better

2nd: verapamil/diltiazem

If β -blocker not tolerated

4th: mavacamten

Myosin inhibitor — echo EF monitoring required

HCM = small cavity, ↑ EF

vs DCM = dilated cavity, ↓ EF

Worsens: Valsalva, standing, dehydration

↓ preload or ↑ contractility → worse

1st: β -blocker ★

Reduces HR + contractility, preserves preload

3rd: disopyramide

Negative inotrope — persistent obstruction

BANNED: nitrates, digoxin, vasodilators

All worsen LVOT obstruction